

Project Plan

Kinetic MHD-PINN

Stellarator energetic-particle–Alfvén-eigenmode digital-twin architecture

Purpose. This document is the single living roadmap for **Kinetic MHD-PINN**. The top-level `README.md`, `project_plan.tex`, and `project_plan.pdf` should be updated whenever the project architecture, interfaces, or module sequence changes.

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Current status. The repository structure, module boundaries, interface contracts, parallel Python/MATLAB architecture, and a 100-note prerequisite curriculum are established. Thirty basics notes are now substantive standalone textbook drafts, version 1.0, while seventy remain scaffolds. Module 4 now contains the first conventional numerical milestone: a reduced self-adjoint two-harmonic shear-Alfvén continuum/eigenmode benchmark with frozen reference outputs, convergence tests, and language-parity criteria. Python execution is verified in the build environment; MATLAB source and tests are complete but have not been executed here. No kinetic drive, orbit-following, transport, or PINN implementation is claimed yet.

1 Project objective

The project will develop a modular digital-twin framework for energetic particles and Alfvén eigenmodes in stellarator burning plasmas. The central scientific objective is to connect magnetic geometry, alpha-particle and neutral-beam distributions, orbit confinement, Alfvén eigenmode structure, energetic-particle kinetic drive, fast-ion redistribution, alpha heating, pressure-profile evolution, wall loads, and diagnostics in one controlled multiphysics architecture.

The project is intentionally hierarchical. Early stages use analytic or reduced geometry, low-dimensional mode models, parameterized kinetic closures, and conservative one-dimensional transport. Higher-fidelity interfaces can later accept VMEC-like equilibria, three-dimensional orbit data, external kinetic-MHD calculations, or experimental diagnostic products without changing the top-level module contracts.

2 Scientific dependency chain

The nominal causal chain is

$$\begin{aligned}
 &\text{geometry/equilibrium} \rightarrow \text{energetic-particle source} \rightarrow \text{orbits and losses} \rightarrow \text{Alfvén spectrum} \\
 &\quad \rightarrow \text{kinetic drive} \rightarrow \text{AE transport} \rightarrow \text{heating and profiles.}
 \end{aligned} \tag{2.1}$$

The most important feedback loops are

$$f_f \rightarrow \gamma_{\text{AE}} \rightarrow A_{\text{AE}} \rightarrow D_{\text{AE}} \rightarrow f_f, \quad (2.2)$$

and

$$f_f \rightarrow P_\alpha \rightarrow (T_i, T_e, p, \rho) \rightarrow (v_A, \omega_{\text{AE}}, S_\alpha) \rightarrow f_f. \quad (2.3)$$

3 Educational physics-note architecture

The repository is also intended to function as a computational-plasma-physics course. In addition to the eleven advanced module notes, `docs/physics_notes/basics` contains 100 stable, numbered LaTeX note scaffolds organized into twelve prerequisite parts: mathematical foundations, electromagnetism, elementary plasma physics, MHD, plasma waves, kinetic theory, fusion physics, stellarator foundations, computational methods, multiphysics coupling, physics-informed machine learning, and graph learning.

At this milestone, thirty basics notes have been expanded into substantive standalone textbook drafts, version 1.0. Batch 1 contains Basics 1, 2, 3, 5, 6, 7, 11, 12, 13, and 17. Batch 2 contains Basics 18, 19, 21, 22, 24, 25, 26, 29, 31, and 32. Batch 3 contains Basics 34, 35, 36, 37, 39, 40, 41, 42, 44, and 48, extending the path from ideal-MHD Alfvén waves through continuum and gap-mode physics into resonance, phase space, guiding-center dynamics, and energetic-particle distributions. Each draft contains derivations, physical interpretation, computational laboratory guidance, exercises, a symbol table, and references. The remaining seventy notes are still explicitly labeled curriculum scaffolds. Every basics note is maintained as a separate document; the project does not create a combined basics volume. The master course map is compiled as `docs/physics_notes/basics/basics_index.pdf`, while `docs/physics_notes/basics/curriculum_manifest.json` provides a machine-readable curriculum contract. Advanced module notes progressively add prerequisite tables using stable basics-note numbers and filenames. Module 4 cites Basics 29, 32, 34–37, 69, and 73. Module 5 now cites Basics 32, 34, 35, 37, 39–42, 44, and 48.

4 Module roadmap

No.	Module	Purpose
1	Stellarator Equilibrium and Magnetic Geometry	Represent the magnetic configuration, flux surfaces, rotational transform, pressure profile, and equilibrium quantities required by every downstream module.
2	Energetic-Particle Sources and Distribution Functions	Construct alpha-particle and neutral-beam source terms and reduced energetic-particle distributions in energy, pitch, radius, and time.

3	Guiding-Center Orbits, Confinement, and Wall Losses	Follow energetic-particle guiding centers in three-dimensional fields, classify orbits, estimate prompt and delayed losses, and map lost-particle energy to the wall.
4	Alfvén Continuum and Fluid-MHD Eigenmodes	Compute the shear-Alfvén continuum, spectral gaps, mode frequencies, and reduced eigenfunctions before energetic-particle kinetic corrections are added.
5	Kinetic Energetic-Particle Drive and Damping	Evaluate resonant energetic-particle drive and reduced damping contributions to determine whether each Alfvén eigenmode grows or decays.
6	Alfvén-Eigenmode-Induced Fast-Ion Transport	Translate unstable mode amplitudes and resonance structure into energetic-particle redistribution, radial diffusion, convection, and loss enhancement.
7	Alpha Heating and Thermal-Profile Evolution	Convert confined energetic-particle power into electron/ion heating and evolve reduced density, temperature, pressure, and Alfvén-speed profiles.
8	Coupled Kinetic-MHD Multiphysics Integration	Orchestrate the equilibrium, fast-ion, eigenmode, transport, heating, and wall-loss modules with explicit interface contracts and controlled feedback loops.
9	PINN and Neural-Operator Surrogates	Develop and compare PINNs, causal PINNs, neural operators, and structure-preserving surrogates for selected expensive forward and inverse maps.
10	Synthetic Diagnostics and Physics-Informed Inverse Modeling	Generate diagnostic signals from simulated states and reconstruct energetic-particle distributions, mode amplitudes, or profile changes from incomplete observations.
11	GNN Physical Dependency Discovery and Coupling Optimization	Infer directed physical-variable dependencies, feedback groups, state-dependent coupling strengths, and candidate numerical exchange sequences from synthetic interventions.

5 Module descriptions and interface anchors

5.1 Module 1: Stellarator Equilibrium and Magnetic Geometry

Purpose. Represent the magnetic configuration, flux surfaces, rotational transform, pressure profile, and equilibrium quantities required by every downstream module.

Primary input. coil or boundary parameters, pressure profile $p(\psi)$, current profile, and equilibrium controls.

Primary output. $\mathbf{B}(\mathbf{x})$, $\psi(\mathbf{x})$, $\iota(\psi)$, metric/Jacobian data, and flux-surface geometry.

Mathematical anchor.

$$\nabla p = \mathbf{J} \times \mathbf{B}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}, \quad \nabla \cdot \mathbf{B} = 0. \quad (5.1)$$

Surrogate opportunity. Physics-informed equilibrium surrogate or neural operator over a family of boundary shapes and profiles.

5.2 Module 2: Energetic-Particle Sources and Distribution Functions

Purpose. Construct alpha-particle and neutral-beam source terms and reduced energetic-particle distributions in energy, pitch, radius, and time.

Primary input. equilibrium geometry, n_D , n_T , T_i , beam parameters, and source normalization.

Primary output. $S_f(\psi, E, \lambda, t)$, $f_f(\psi, E, \lambda, t)$, birth power, and moments of the fast-ion population.

Mathematical anchor.

$$\frac{\partial f_f}{\partial t} + \dot{\mathbf{Z}} \cdot \nabla_{\mathbf{Z}} f_f = C[f_f] + S_f - L_f. \quad (5.2)$$

Surrogate opportunity. Physics-informed or operator-learning surrogate for reduced Fokker-Planck evolution and source-to-distribution maps.

5.3 Module 3: Guiding-Center Orbits, Confinement, and Wall Losses

Purpose. Follow energetic-particle guiding centers in three-dimensional fields, classify orbits, estimate prompt and delayed losses, and map lost-particle energy to the wall.

Primary input. $\mathbf{B}(\mathbf{x})$, electric field if retained, particle birth coordinates, energy, pitch, and collision model.

Primary output. orbit histories, confinement fraction, loss time, loss phase space, and $q_{\text{wall}}(\theta, \phi, t)$.

Mathematical anchor.

$$\dot{\mathbf{R}} = v_{\parallel} \mathbf{b} + \mathbf{v}_E + \mathbf{v}_{\nabla B} + \mathbf{v}_c, \quad \dot{v}_{\parallel} = -\frac{\mu}{m} \mathbf{b} \cdot \nabla B + \frac{q}{m} E_{\parallel}. \quad (5.3)$$

Surrogate opportunity. Structure-preserving neural surrogate for orbit maps, invariants, confinement classification, and wall-loss probability.

5.4 Module 4: Alfvén Continuum and Fluid-MHD Eigenmodes

Purpose. Compute the shear-Alfvén continuum, spectral gaps, mode frequencies, and reduced eigenfunctions before energetic-particle kinetic corrections are added.

Primary input. equilibrium geometry, mass density, pressure, rotational transform, and harmonic truncation.

Primary output. ω_j , ξ_j , continuum branches, gap locations, and ideal-MHD energy terms.

Mathematical anchor.

$$\mathcal{L}_{\text{MHD}}[\xi_j] = \omega_j^2 \mathcal{M}[\xi_j]. \quad (5.4)$$

Implemented F0 milestone. A reduced two-harmonic avoided-crossing operator is solved with symmetric finite-volume matrices in Python and MATLAB. The next surrogate milestone is an eigenvalue PINN with PDE residual, boundary conditions, normalization, orthogonality, and Rayleigh-quotient consistency.

5.5 Module 5: Kinetic Energetic-Particle Drive and Damping

Purpose. Evaluate resonant energetic-particle drive and reduced damping contributions to determine whether each Alfvén eigenmode grows or decays.

Primary input. mode structure, mode frequency, equilibrium, orbit frequencies, and f_f .

Primary output. γ_{EP} , damping terms, net γ_j , resonance maps, and stability labels.

Mathematical anchor.

$$\gamma_j = \gamma_{\text{EP},j} - \gamma_{\text{cont},j} - \gamma_{\text{Landau},j} - \gamma_{\text{FLR},j} - \gamma_{\text{other},j}. \quad (5.5)$$

Surrogate opportunity. Physics-regularized surrogate mapping equilibrium and energetic-particle distributions to mode frequency, growth rate, and uncertainty.

5.6 Module 6: Alfvén-Eigenmode-Induced Fast-Ion Transport

Purpose. Translate unstable mode amplitudes and resonance structure into energetic-particle redistribution, radial diffusion, convection, and loss enhancement.

Primary input. f_f , ω_j , γ_j , mode amplitudes, resonance functions, and transport closure parameters.

Primary output. D_{AE} , V_{AE} , updated f_f , mode saturation histories, and redistributed loss power.

Mathematical anchor.

$$\frac{\partial f_f}{\partial t} = \frac{1}{V'} \frac{\partial}{\partial \psi} \left[V' D_{\text{AE}} \frac{\partial f_f}{\partial \psi} - V' V_{\text{AE}} f_f \right] + S_f - C[f_f]. \quad (5.6)$$

Surrogate opportunity. Causal PINN or neural operator for stiff transport with conservation, positivity, and resonance-informed losses.

5.7 Module 7: Alpha Heating and Thermal-Profile Evolution

Purpose. Convert confined energetic-particle power into electron/ion heating and evolve reduced density, temperature, pressure, and Alfvén-speed profiles.

Primary input. deposited alpha/NBI power, loss power, transport coefficients, radiation losses, and initial profiles.

Primary output. $T_i(\psi, t)$, $T_e(\psi, t)$, $p(\psi, t)$, $\rho(\psi, t)$, $v_A(\psi, t)$, and fusion-source updates.

Mathematical anchor.

$$\frac{3}{2} \frac{\partial(n_s T_s)}{\partial t} = -\frac{1}{V_I} \frac{\partial}{\partial \psi} (V_I Q_s) + P_{\alpha \rightarrow s} + P_{\text{NBI} \rightarrow s} - P_{\text{loss}, s}. \quad (5.7)$$

Surrogate opportunity. Neural operator for real-time profile evolution constrained by energy conservation and transport equations.

5.8 Module 8: Coupled Kinetic-MHD Multiphysics Integration

Purpose. Orchestrate the equilibrium, fast-ion, eigenmode, transport, heating, and wall-loss modules with explicit interface contracts and controlled feedback loops.

Primary input. all module states, coupling tolerances, exchange cadence, relaxation parameters, and conservation diagnostics.

Primary output. self-consistent coupled trajectories, convergence histories, conserved-quantity errors, and solver-cost metrics.

Mathematical anchor.

$$\mathbf{R}(\mathbf{U}) = \mathbf{0}, \quad \mathbf{U} = [\mathbf{B}, f_f, \boldsymbol{\xi}, \omega, \gamma, A, D_{\text{AE}}, T_i, T_e, p]^T. \quad (5.8)$$

Surrogate opportunity. Hybrid solver combining first-principles modules with learned closures while preserving hard conservation and interface constraints.

5.9 Module 9: PINN and Neural-Operator Surrogates

Purpose. Develop and compare PINNs, causal PINNs, neural operators, and structure-preserving surrogates for selected expensive forward and inverse maps.

Primary input. high-fidelity or reduced-model datasets, governing residuals, boundary conditions, conservation laws, and uncertainty targets.

Primary output. trained surrogates, error budgets, calibration metrics, speedups, and admissibility diagnostics.

Mathematical anchor.

$$\mathcal{L} = \lambda_{\text{data}}\mathcal{L}_{\text{data}} + \lambda_{\text{PDE}}\mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}}\mathcal{L}_{\text{BC}} + \lambda_{\text{cons}}\mathcal{L}_{\text{cons}}. \quad (5.9)$$

Surrogate opportunity. This module is the cross-cutting surrogate layer and defines model-selection criteria for each physical subproblem.

5.10 Module 10: Synthetic Diagnostics and Physics-Informed Inverse Modeling

Purpose. Generate diagnostic signals from simulated states and reconstruct energetic-particle distributions, mode amplitudes, or profile changes from incomplete observations.

Primary input. coupled plasma state, diagnostic geometry, response functions, noise model, and observation cadence.

Primary output. synthetic magnetic, neutron, fast-ion-loss, and heat-load signals plus reconstructed latent states and uncertainties.

Mathematical anchor.

$$y_k(t) = \mathcal{H}_k[\mathbf{U}(t)] + \epsilon_k(t). \quad (5.10)$$

Surrogate opportunity. Physics-informed inverse model combining diagnostic likelihoods with transport, eigenmode, and conservation residuals.

5.11 Module 11: GNN Physical Dependency Discovery and Coupling Optimization

Purpose. Infer directed physical-variable dependencies, feedback groups, state-dependent coupling strengths, and candidate numerical exchange sequences from synthetic interventions.

Primary input. multimodule trajectories, intervention labels, mechanism switches, time lags, and candidate graph priors.

Primary output. directed typed graphs, edge strengths and lags, strongly connected feedback groups, and ranked coupling strategies.

Mathematical anchor.

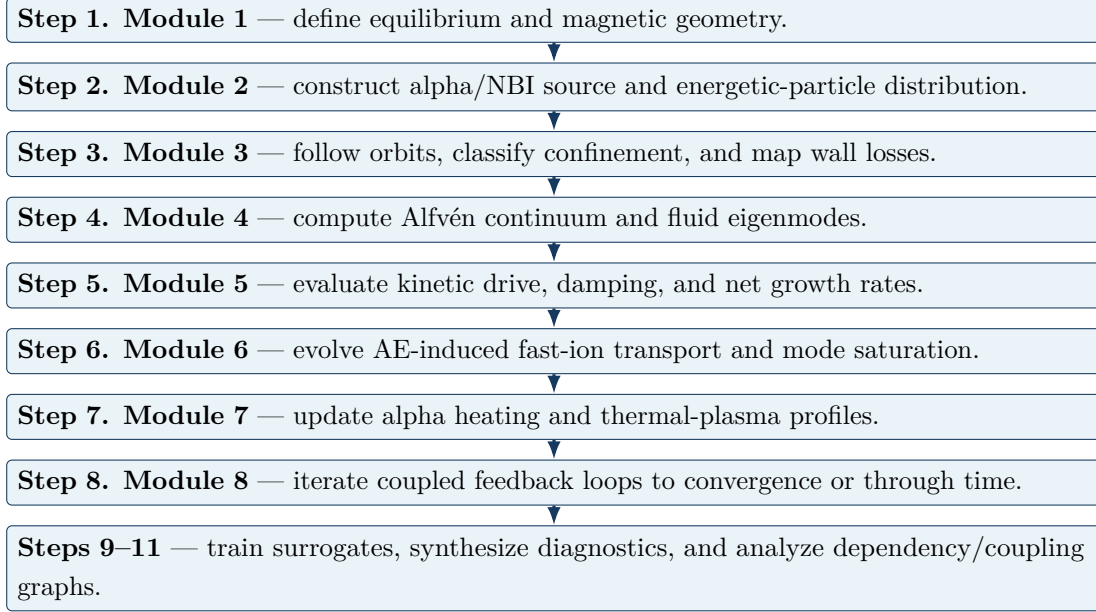
$$\hat{G} = \arg \min_G (\mathcal{L}_{\text{trajectory}} + \lambda_{\text{phys}}\mathcal{L}_{\text{admissibility}} + \lambda_{\text{sparse}}\|G\|_1). \quad (5.11)$$

Surrogate opportunity. Graph learning complements rather than replaces physical reasoning: causality, conservation, and timescale constraints remain hard admissibility filters.

6 Fidelity ladder

Level	Model scope	Purpose
F0	Analytic cylindrical or large-aspect-ratio toroidal benchmark	Verify signs, units, limiting behavior, eigenvalue normalization, and conservation before stellarator complexity is introduced.
F1	Reduced radial TAE/energetic-particle model	Establish the first publishable/interview-ready kinetic-MHD feedback loop with conventional solvers and PINN comparison.
F2	Analytic three-dimensional stellarator field plus guiding-center ensemble	Add non-axisymmetric orbit confinement, prompt loss, resonance maps, and wall strike distributions.
F3	Imported numerical equilibrium and external-code interfaces	Accept equilibrium, orbit, eigenmode, or diagnostic products from specialized tools while maintaining project-owned data contracts.
F4	Calibrated integrated surrogate/digital-twin layer	Support rapid parameter scans, uncertainty propagation, profile prediction, and diagnostic inversion with hard physical checks.

7 Nominal run-order workflow



8 Coupling strategy

The coupling sequence is not treated as an unconstrained machine-learning problem. Physical causality, conservation, and timescale separation define the admissible search space. Candidate numerical strategies include:

- one-way initialization followed by staggered Picard iteration;
- block Gauss–Seidel exchange for tightly coupled feedback groups;
- under-relaxation or Anderson acceleration;
- waveform relaxation for modules with different natural time steps;
- quasi-Newton coupling for the f_f –AE growth–transport loop;
- conservative remapping between phase-space, radial-profile, and wall meshes;
- learned closures only when a hard residual or conservation check remains available.

A generic coupling objective is

$$\mathcal{J}_{\text{coupling}} = w_E \epsilon_E + w_N \epsilon_N + w_B \epsilon_{\nabla \cdot B} + w_R \|\mathbf{R}\| + w_I N_{\text{iter}} + w_t t_{\text{wall}}. \quad (8.1)$$

9 Core state and data interfaces

State product	Produced by	Primary consumers
Equilibrium field and flux geometry	Module 1	Modules 2–7
Energetic-particle source/distribution	Module 2	Modules 3, 5, 6, and 7
Orbit frequencies and wall losses	Module 3	Modules 5, 6, 7, and 10
Mode frequency and structure	Module 4	Modules 5, 6, 9, and 10
Growth/damping rates and resonances	Module 5	Modules 6, 8, 9, and 10
Transport coefficients and redistributed f_f	Module 6	Modules 2, 3, 7, and 8
Thermal-plasma profiles	Module 7	Modules 1, 2, 4, 5, and 8
Coupled trajectories and residuals	Module 8	Modules 9–11
Surrogate predictions and uncertainty	Module 9	Module 8 and rapid-scan workflows
Synthetic diagnostic observations	Module 10	Inverse-model and validation studies
Dependency graph and coupling recommendation	Module 11	Module 8 solver-design studies

10 Parallel Python and MATLAB development

The project maintains two independent implementations of the same physical architecture. The language trees shall share module numbering, governing equations, normalization conventions, configuration inputs, state-product names, verification cases, and output definitions. The MATLAB tree uses package directories beginning with `+`, which are the language-specific equivalent of Python package directories.

The implementations should not become line-by-line translations. Independent discretizations and language ecosystems provide a useful cross-check, but numerical agreement alone is not validation. Each implementation must separately satisfy dimensional checks, analytic limits, conservation diagnostics, convergence studies, and external reference comparisons. Shared data and configuration files remain at the repository root to prevent silent divergence of physical assumptions.

11 Repository file map

Path	Role
README.md	Repository summary, current status, and entry point.
project_plan.tex/.pdf	Single living roadmap for architecture, coupling, interfaces, and implementation order.
docs/shared/project_style.tex	Shared LaTeX style inherited from the Radiation Effects project and renamed for this project.
docs/physics_notes/	Advanced module derivations and physics notes.
docs/physics_notes/basics/	One hundred stable prerequisite-note sources: thirty standalone textbook drafts, seventy scaffolds, a compiled status-aware index, and a machine-readable manifest.
docs/implementation_notes/	Module software contracts, algorithms, verification cases, and interface notes.
python/src/kinetic_mhd_pinn/	Python package arranged by physical module plus common state and coupling utilities.
matlab/src/+kinetic_mhd_pinn/	MATLAB package mirror with the same module contracts and language-appropriate classes/functions.
python/cases/,matlab/cases/	Reproducible benchmark and demonstration drivers in both languages.
python/tests/,matlab/tests/	Contract, verification, cross-language, and later regression tests.
configs/	Version-controlled run configurations.
data/raw,processed,reference/	Input, transformed, and benchmark/reference datasets.

12 Verification hierarchy

Every implemented module should pass four levels of evidence:

1. **Dimensional and sign checks:** units, normalizations, conserved quantities, and direction conventions.
2. **Analytic limits:** uniform fields, no-drive limits, zero-gradient distributions, decoupled modes, and manufactured solutions.
3. **Cross-solver comparison:** independent Python and MATLAB implementations, independent discretizations, conventional solver versus PINN, or reduced model versus external reference data.

4. **Coupled diagnostics:** particle and energy budgets, mode-energy exchange, profile consistency, convergence, and sensitivity to exchange cadence.

13 Recommended coding order

1. **Completed conventional stage:** implement Module 4 as a reduced two-harmonic TAE-like eigenvalue benchmark independently in Python and MATLAB, with frozen reference outputs and convergence tests.
2. Implement the Module 4 eigenvalue PINN and validate it against the conventional matrix solver.
3. Add Module 2 as a parameterized slowing-down energetic-particle distribution.
4. Implement a reduced Module 5 kinetic-drive closure and produce stability maps.
5. Add Module 6 one-dimensional quasilinear redistribution and mode-amplitude saturation.
6. Couple Module 7 profile/heating evolution and compare explicit, staggered, and accelerated coupling.
7. Introduce Module 1/3 stellarator geometry and guiding-center orbits after the reduced AE loop is verified.
8. Add Modules 9–11 only after credible conventional-solver datasets exist.

14 Milestone status and forward path

Status	Milestone	Evidence or next acceptance criterion
Completed	Module 1 textbook physics note, version 1.0	Comprehensive introductory-to-advanced treatment of stellarator equilibrium, flux surfaces, rotational transform, magnetic coordinates, three-dimensional Fourier geometry, equilibrium-solver concepts, output contracts, and verification; no three-dimensional solver is claimed.

Completed	Reduced two-harmonic Alfvén continuum/eigenmode benchmark	Python reference execution, matched MATLAB source, frozen CSV/JSON data products, continuum and eigenmode figures, scaling tests, decoupled-limit checks, and grid refinement.
Next	Module 4 eigenvalue PINN	Recover selected eigenfrequencies and eigenfunctions within prescribed tolerances while satisfying differential residuals, boundary conditions, normalization, and orthogonality.
Planned	Kinetic-drive stability map	Add a parameterized energetic-particle distribution, a reduced resonance/drive closure, net growth-rate scans, and uncertainty-aware surrogate comparison.
Planned	AE/fast-ion feedback loop	Add mode saturation, quasi-linear redistribution, energy/particle diagnostics, and explicit versus staggered or accelerated coupling comparisons.

15 Current baseline boundary

This baseline contains a completed Module 1 educational physics note and the first validated conventional numerical capability, but its executable scope remains intentionally reduced. It does not yet solve a three-dimensional stellarator equilibrium, guiding-center orbit populations, energetic-particle drive and damping, nonlinear fast-ion transport, alpha-heating feedback, synthetic diagnostics, or any PINN. The present Module 4 operator is a controlled two-harmonic radial benchmark for solver verification and future surrogate validation; it must not be represented as a predictive device-scale kinetic-MHD calculation. The next repository milestone is the Module 4 eigenvalue PINN, retained alongside—not in place of—the conventional matrix solver.